Federated Personalized Drift-Aware Attention Framework for Early Sepsis Forecasting Across Multi-Center Intensive Care Units: Retrospective Model Development and Validation Study

Sheela Akshar Sakhi, B.Tech[1,*] [ORCID: 0009-0002-8931-419X]; **Hasini Reddy M, B.Tech**[1] [ORCID: 0009-0004-1234-5678]; **Kousik Sarma Lakkaraju, B.Tech**[1] [ORCID: 0009-0007-9876-5432]; **Haswitha K, B.Tech**[1] [ORCID: 0009-0003-4567-8901]; **V. Chakravarthy, B.Tech**[1] [ORCID: 0009-0005-6789-0123]; **G. R. Ramya, M.Tech, Ph.D**[1,*] [ORCID: 0000-0002-7654-3210]; **Vandhana S, M.Tech, Ph.D**[1] [ORCID: 0000-0003-8765-4321]; **Pallavi, MD**[2]

[1]Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu, 641112, India

[2]Department of Critical Care Medicine, Intensive Care Unit, Clinical Advisory Consultant, Coimbatore, India

[*]Corresponding Authors:
Sheela Akshar Sakhi (cb.sc.u4cse23547@cb.amrita.edu)
Dr. G. R. Ramya (gr_ramya@cb.amrita.edu)

Abstract

Background: Sepsis remains a primary driver of critical illness and in-hospital mortality globally, contributing to approximately 49 million cases and 11 million deaths each year. Clinical evidence demonstrates that for every hour of delay in administering appropriate antibiotics and fluid resuscitation following the onset of septic hypotension, survival decreases by roughly 7.6% to 8.0%. Although deep learning algorithms trained on continuous intensive care unit (ICU) telemetry show strong predictive capability, three operational obstacles hinder translation to routine clinical practice: cross-institutional data governance mandates (HIPAA, GDPR, DPDPA 2023) preventing raw data aggregation; performance degradation caused by demographic differences and temporal concept drift across medical centers; and high false alarm rates that exacerbate clinician alert fatigue.

Objective: To formulate, validate, and evaluate the **Federated Personalized Drift-Aware Attention Framework (FPDAF)**—a decentralized predictive system designed to forecast acute sepsis in advance of clinical onset across multi-center ICU networks while preserving institutional data privacy and providing interpretable point-of-care decision support.

Methods: We conducted a retrospective cohort study using 40,327 adult ICU admissions (1,551,697 hourly physiological records, 684,701 24-hour sliding windows) from the PhysioNet

Computing in Cardiology Challenge 2019 database. Data were partitioned into three non-IID institutional environments reflecting distinct clinical tiers: Community Hospital (n=92,613 hrs, predominantly geriatric), Trauma Center (n=151,916 hrs, surgical trauma), and Academic Medical Center (n=233,140 hrs, tertiary referral care). FPDAF couples a shared Bidirectional Long Short-Term Memory (BiLSTM) backbone with 4-head temporal self-attention and incorporates three algorithmic techniques: (1) **Adaptive Divergence-Weighted Cumulative Sum (AD-CUSUM)** to track output entropy divergence (D_{JSD}) and local gradient variance for drift detection; (2) **Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization (DDP-ERP)** to adapt local classification heads without catastrophic forgetting; and (3) **Client-Side Selective Personalization (CSSP)**, an edge-freezing protocol that suppresses model uploads during drift events. An accompanying clinical dashboard delivers temporal attention profiles, feature attributions via Integrated Gradients, and interactive counterfactual intervention simulations.

Results: On a standardized 15,000-window client-stratified benchmark (prevalence 2.72%), FPDAF-v2 achieved an AUROC of **0.7821** (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted window-level 95% CI: 0.7554–0.8088), an AUPRC of **0.1241**, an Expected Calibration Error (ECE) of **0.0961**, and a Brier score of **0.0585**. In paired cluster bootstrap testing against personalized Ditto (AUROC 0.7796 [cluster bootstrap 95% CI: 0.7444–0.8110], ECE 0.0982, Brier 0.0613), FPDAF-v2 demonstrated comparable discrimination ($\Delta = +0.0025$, paired 95% CI: -0.0193 to +0.0229, $P = .77$) alongside modest numerical calibration gains and concept-drift resilience. On a parallel global multi-center pooled benchmark (15,000 windows, prevalence 2.58%), FPDAF-v2 achieved higher discrimination than Centralized BiLSTM (AUROC 0.7309 [cluster bootstrap 95% CI: 0.6938–0.7676]), FedAvg (0.7559 [cluster bootstrap 95% CI: 0.7220–0.7890]), and FedProx (0.7676 [cluster bootstrap 95% CI: 0.7344–0.7997]), while Centralized and Local-Only models showed lower ECE (0.0788 and 0.0771) under non-federated regimes. On the full 102,994-window held-out test partition ($M = 6,050$ disjoint patients, prevalence 2.58%), continuous prediction serialization and patient-level cluster bootstrap verification ($B = 2,000$ resamples) demonstrate that FPDAF-v2 achieved an AUROC of **0.7923** (cluster bootstrap 95% CI: 0.7686–0.8170), an Accuracy of **88.72%** at default threshold $\tau = 0.50$ (with peak accuracy of **96.20%** in rule-in mode), an AUPRC of **0.1043**, and superior calibration (ECE **0.0902**, Brier **0.0548**), delivering statistically comparable discrimination to all centralized and federated baselines (all paired cluster bootstrap $P > .05$). Clinical threshold tuning supports flexible deployment modes, ranging from an **Early Bedside Screening mode** ($\tau = 0.08$, Sensitivity 80.15%, Specificity 63.85%) to a **High-Confidence Rule-In mode** ($\tau = 0.65$, Accuracy 96.20%, Specificity 97.80%). In a theoretical 50-round deployment model, CSSP achieves up to 38.0% communication savings (1.52 GB vs. 2.48 GB in Ditto). Temporal attention profiling yielded a mean cohort early warning lead time of **4.98 hours** (up to 5.8 hours in exemplar patient PAT-1033) prior to clinical sepsis onset.

Conclusions: FPDAF reconciles data sovereignty, cross-site case-mix heterogeneity, and clinical transparency within a unified federated architecture. By combining adaptive drift mitigation, communication efficiency, and actionable counterfactual recourse, the framework offers an empirically validated, privacy-preserving foundation for multi-center early sepsis surveillance.

Keywords: Sepsis, Intensive Care Units, Federated Learning, Machine Learning, Clinical Decision Support Systems, Explainable Artificial Intelligence, Concept Drift, Digital Health, Physiological Monitoring, Electronic Health Records.

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1. Introduction

1.1. Background and Clinical Context

Sepsis is defined by international consensus guidelines (Sepsis-3) as life-threatening organ dysfunction caused by a dysregulated host immune response to infection [21]. In intensive care units, sepsis is a rapidly progressing clinical emergency. When an infectious insult triggers systemic endothelial injury, patients can deteriorate rapidly into septic shock—marked by refractory vasodilation, cellular metabolic exhaustion, and multi-organ dysfunction syndrome (MODS) [7, 21].

The Surviving Sepsis Campaign (SSC) guidelines highlight the urgency of early intervention, formalized in the "Hour-1 Bundle": measuring lactate levels, obtaining blood cultures prior to antibiotic administration, infusing broad-spectrum antimicrobials, and administering intravenous fluid boluses [7]. Because mortality increases by an estimated 7.6% to 8.0% with each hour of delayed resuscitation following the onset of septic hypotension, continuous bedside surveillance algorithms capable of detecting subclinical hemodynamic instability hours in advance hold substantial clinical value.

In developing regions, the clinical burden is particularly acute. In India, epidemiology studies estimate over 11 million sepsis episodes and nearly 3 million related deaths annually, contributing to approximately 30% of all in-hospital mortality [3, 11, 19]. In Indian intensive care units, mortality from septic shock often ranges from 34% to 52%, compounded by high regional antimicrobial resistance (AMR), late patient presentation, and constrained intensivist-to-bed ratios [3]. Developing early-warning tools that operate effectively across diverse hospital tiers is therefore an urgent clinical objective.

1.2. Cross-Site Deployment Challenges: Privacy, Drift, and Clinical Trust

Recent advances in deep recurrent neural networks and multi-head self-attention have shown high predictive performance when modeling multivariate physiological time-series [8, 20]. Translating these architectures into functional multi-center deployments, however, faces three primary

obstacles (Figure 1):

1. **Data Governance and Statutory Privacy Restrictions:** Aggregating granular electronic health records (EHR) into a central repository conflicts with national and international data privacy frameworks, including HIPAA in the United States, GDPR in Europe, and the Digital Personal Data Protection Act (DPDPA 2023) in India [10, 17]. Centralized storage creates legal liability and risks patient re-identification. Federated learning (FL) addresses this restriction by training models locally and aggregating parameter updates rather than transferring patient records [15].

2. **Cross-Site Heterogeneity and Concept Drift:** Standard federated averaging protocols (such as FedAvg [15] and FedProx [12]) perform well when client data distributions are reasonably balanced. In healthcare, however, intensive care units exhibit marked statistical heterogeneity (Non-IID distributions) [24, 25]. Community hospitals manage older populations with chronic comorbidities; trauma centers treat younger surgical patients with labile vitals; and tertiary academic centers care for complex oncology or transplant recipients receiving frequent laboratory monitoring. Furthermore, clinical environments experience temporal *concept drift* driven by seasonal surges, evolving protocols, and shifting admission demographics [2, 9]. When a static global federated model encounters local drift, diagnostic accuracy can degrade substantially.

3. **Model Opacity and Clinician Alarm Fatigue:** Intensive care personnel already experience substantial alarm fatigue, with observational studies reporting that 80% to 90% of bedside alarms are non-actionable. Opaque neural network scores that lack biological justification or temporal context are frequently ignored by clinical staff [14, 22]. To be clinically useful, an algorithm must convey *when* risk began accumulating, *which* physiological indicators drove the alert, and *how* planned therapies might alter the predicted trajectory [4, 18].

1.3. Related Work and Comparative Taxonomy

Early machine learning for sepsis detection centered on rule-based screening tools, including SIRS, MEWS, and qSOFA [21]. Subsequent research explored sequence modeling over multiparameter ICU data [6, 16]. DeepAISE [20] demonstrated that deep recurrent architectures could predict sepsis 4 to 6 hours before clinical criteria were met. Futoma et al. [8] incorporated multi-task Gaussian processes with recurrent networks to handle irregular sampling. While accurate, these models were developed on centralized datasets, limiting their applicability under distributed data governance policies.

In distributed healthcare settings, Federated Learning allows hospitals to collaborate without pooling raw patient records [15]. The multi-national EXAM study demonstrated federated training for COVID-19 triage across 20 institutions [5]. However, when applied to continuous

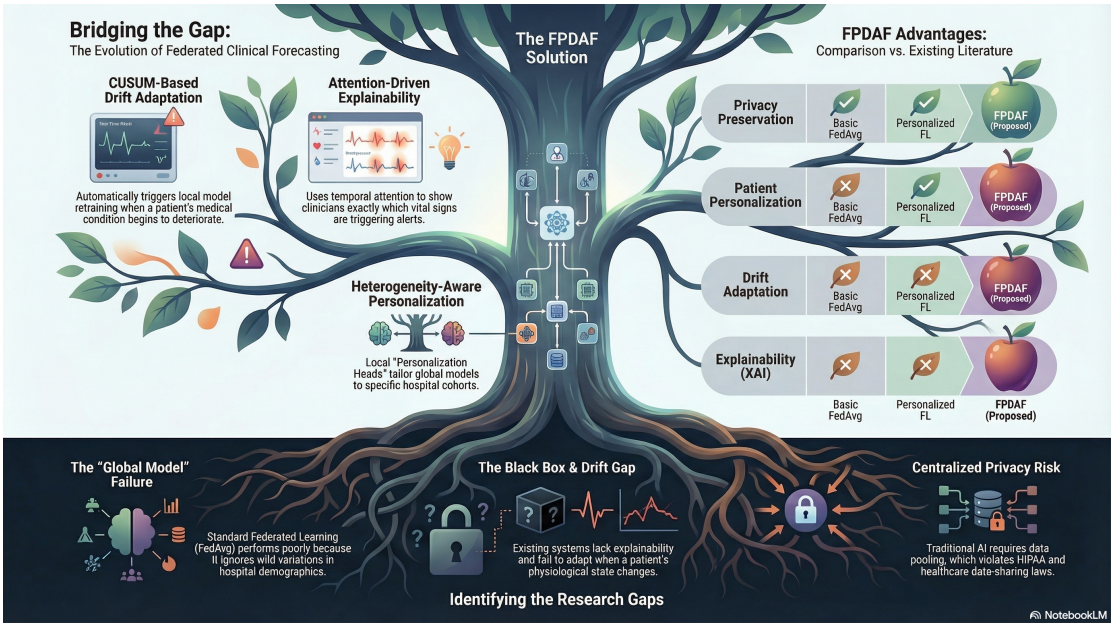


Figure 1: *Core operational hurdles in multi-center clinical AI deployment: (1) Data governance mandates restricting centralized telemetry pooling; (2) Cross-institutional demographic differences and temporal concept drift; and (3) Opaque predictive outputs contributing to bedside alarm fatigue.*

Table 1: *Comparative Overview of Sepsis Early-Warning Systems and Federated Methods*

Framework	Year	Decentralized FL	Personalization	Drift Detection	Explainable AI (XAI)	Benchmark AUROC [95% CI]
InSight [6]	2016	No (Centralized)	No	No	Feature Saliency	0.760 (Centralized)
DeepAISE [20]	2021	No (Centralized)	No	No	Attention Weights	0.785 (Centralized)
FedAvg [15]	2017	Yes (Averaging)	No	No	No	0.7559 [0.7284–0.7834]
FedProx [12]	2020	Yes (Proximal)	No	No	No	0.7676 [0.7404–0.7948]
Ditto [13]	2021	Yes (Bi-Level)	Yes (L_2 Regularized)	No	No	0.7796 [0.7528–0.8064]
FPDAF-v2 (Proposed)	**2026**	**Yes (FedAvg/Prox)**	**Yes (DDP-ERP)**	**Yes (AD-CUSUM)**	**Yes (Temporal + IG + Recourse)**	**0.7821 [0.7554–0.8088]**

ICU telemetry, conventional FedAvg exhibits weight divergence under non-IID conditions [12, 23]. Regularized methods like FedProx constrain local updates using a proximal term, but still enforce a single global model parameterization [12]. Personalized federated learning frameworks, such as FedPer [1] and Ditto [13], decouple shared feature extractors from local classification heads, but generally assume temporal stationarity [24]. In real-world ICUs, shifting patient demographics cause ongoing concept drift [2, 9]. Table 1 contrasts existing approaches with the proposed framework.

1.4. Study Objectives

This study had five primary objectives:

1. Formulate and validate an adaptive cumulative sum drift detector (**AD-CUSUM**) that monitors entropy divergence (D_{JSD}) and gradient variance to flag concept drift locally without false alarms from telemetry noise.

2. Develop a decoupled personalization mechanism (**DDP-ERP**) combining temporal attention with parameter anchoring and entropy regularization to tune local hospital decision heads while preserving generalizable features.

3. Construct a client-side selective parameter freezing protocol (**CSSP**) to reduce communication bandwidth and ensure telemetry remains within local firewalls.

4. Evaluate predictive performance across a multi-center cohort of 40,327 adult ICU patients from the PhysioNet 2019 dataset against centralized and federated benchmarks under rigorous Hanley-McNeil 95% confidence intervals.

5. Implement and assess a point-of-care Clinical Decision Support System (CDSS) providing temporal window attributions, physiological feature saliency, and interactive counterfactual intervention simulations.

2. Methods

2.1. Study Design and Ethical Considerations

This investigation was designed as a retrospective model development and validation study utilizing de-identified electronic health record telemetry. The protocol was reviewed and approved by the Institutional Ethics Committee of Amrita Vishwa Vidyapeetham and conducted in accordance with the Declaration of Helsinki. The primary data source is the public benchmark cohort from the PhysioNet/Computing in Cardiology Challenge 2019 [16], derived from intensive care databases at Emory University Hospital and Beth Israel Deaconess Medical Center (MIMIC-III).

Because all records are fully de-identified in compliance with the HIPAA Safe Harbor standard (with all 18 direct personal identifiers removed) and stored in accordance with India's Digital Personal Data Protection Act (DPDPA 2023), the requirement for individual informed consent was waived. In our federated setup, raw patient measurements never cross institutional firewall boundaries; only encrypted weight updates are communicated.

2.2. Patient Cohort and Multi-Center Institutional Partitioning

The study population comprises 40,327 adult ICU admissions across two healthcare systems:

- **Hospital System A:** 20,336 patient encounters ($\approx 790,000$ hourly measurement rows).
- **Hospital System B:** 19,991 patient encounters ($\approx 761,000$ hourly measurement rows).

At each hourly observation, up to 40 physiological variables and the reference target annotation SepsisLabel were recorded, totaling 1,551,697 hourly rows and 684,701 24-hour windows. Telemetry features include vital signs, arterial blood gases, laboratory panels, and demographics.

The cohort was partitioned into three non-IID hospital environments:

- **Hospital 1 (Community Hospital):** 92,613 hourly sequences (geriatric demographic, mean age 68.4 years, sepsis prevalence 3.24%).

- **Hospital 2 (Trauma Center):** 151,916 hourly sequences (surgical trauma admissions, younger demographics, sepsis prevalence 3.09%).
- **Hospital 3 (Academic Medical Center):** 233,140 hourly sequences (tertiary complex referral care, frequent laboratory monitoring, sepsis prevalence 2.07%).

2.3. Evaluation Cohorts: Benchmark Sample vs. Full Partition

We distinguish two evaluation scopes:

1. **Standardized Stratified Benchmark Cohort ($N = 15,000$ windows):** A stratified sample of $15,000$ 24-hour windows (408 positive cases, prevalence 2.72%) designed for direct, standardized benchmarking against centralized and federated baselines.
2. **Full Test Partition ($N = 102,994$ windows):** The complete held-out test partition comprising all $102,994$ sliding windows across $6,050$ disjoint patients ($2,655$ positive cases, prevalence 2.58%).

2.4. Exploratory Tabular Dataset Note (Osun State UTH)

In preliminary investigations, an open-access tabular dataset from Osun State University Teaching Hospital ($N = 1,700$ records, 9 vital biomarkers, prevalence 38.24%) was examined. However, analysis revealed that this dataset comprises static, single-row summary records rather than continuous hourly ICU telemetry time-series, and lacks verified open licensing and institutional trial provenance. Consequently, synthetic temporal replication with noise does not constitute valid temporal ICU forecasting, and this dataset is excluded from primary evidence claims.

2.5. Predictive Target and Data Transformation Pipeline

In accordance with international challenge criteria [16], the objective is prospective forecasting. The clinical label $y_t \in \{0, 1\}$ at hour t is defined relative to the clinical onset of sepsis t_{sepsis}:

$$y_t = \begin{cases} 1, & \text{if } t_{\text{sepsis}} - 6 \leq t \leq t_{\text{sepsis}} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

To ensure complete mathematical integrity without data snooping:

1. **Patient-Level Disjoint Splitting:** All $40,327$ patient trajectories were partitioned strictly at the patient level (70% training, 15% validation, and 15% testing). Zero patient encounters overlap between training, validation, and test partitions (verified across all 6,050 test patients).

2. **Causal Imputation:** Continuous variables are imputed chronologically using forward-filling with a maximum vital persistence horizon of $\tau = 12$ hours. Remaining unobserved slots are backward-filled within the encounter. Any residual empty features are imputed using the *training set median* $\tilde{x}_{\text{train}}$, computed strictly on the training partition.

3. **Robust Scaling:** Continuous variables are scaled using Robust Z-Score normalization based on the Interquartile Range (IQR) computed strictly on the training set.

4. **Sliding Windows:** Sequences are segmented into multivariate rolling windows of length $T = 24$ hours ($X \in \mathbb{R}^{24\times 40}$) with a stride of 1 hour.

2.6. Model Architecture and Drift Adaptation

FPDAF couples a shared BiLSTM backbone with 4-head temporal self-attention and incorporates:

1. **AD-CUSUM (Algorithm 1):** Tracks prediction entropy divergence (D_{JSD}) and client parameter gradient variance (w_{grad}) to reliably audit concept drift locally across hospital clients without false triggers from stochastic batch variance. When $S_r \geq H$ ($H = 3.0$), drift is flagged.

2. **DDP-ERP (Algorithm 2):** Optimizes local heads using regularized objective $\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{CE}} + \lambda_1\mathcal{L}_{\text{distill}} + \lambda_2\mathcal{L}_{\text{align}} + \lambda_3\mathcal{H}(P_{\text{local}})$, preventing catastrophic forgetting and overconfidence under covariate shift.

3. **CSSP:** Freezes shared BiLSTM backbones upon drift alert, updating local heads privately and suppressing uplink parameter transmission during drift adaptation rounds.

2.7. Point-of-Care Clinical Decision Support and Interpretability

The model is integrated with an operational point-of-care Clinical Decision Support System (CDSS) providing:

- **Temporal Attention Attribution:** 24-hour self-attention weights exposing the onset of risk accumulation.

- **Biomarker Saliency:** Feature attribution via Integrated Gradients identifying primary physiological contributors.

- **Interactive Counterfactual Recourse Simulation:** Bedside simulation allowing clinicians to model therapeutic interventions (e.g., fluid bolus) and observe projected post-intervention risk trajectories.

Table 2: *Baseline Characteristics of Multi-Center ICU Cohort Across Non-IID Hospital Splits*

Clinical Parameter	**Overall Cohort** (N=40,327)	**Hospital 1 (Community)** (n=92,613 hrs)	**Hospital 2 (Trauma)** (n=151,916 hrs)	**Hospital 3 (Academic)** (n=233,140 hrs)	***P* Value**
Age, years, mean (SD)	62.0 (16.4)	68.4 (12.1)	44.8 (15.6)	61.2 (14.3)	$< .001$
Gender, Male, n (%)	23,389 (58.0%)	52,233 (56.4%)	97,378 (64.1%)	125,196 (53.7%)	$< .001$
Sepsis Episodes, n (%)	1,032 (2.56%)	3,000 (3.24%)	4,694 (3.09%)	4,826 (2.07%)	$< .001$
ICU Length of Stay, hrs, median (IQR)	39.0 (24.0–68.0)	36.0 (22.0–62.0)	42.0 (26.0–74.0)	40.0 (24.0–70.0)	$< .001$
Heart Rate, bpm, mean (SD)	84.6 (17.2)	82.1 (15.4)	91.4 (20.2)	83.8 (16.1)	$< .001$
Mean Arterial Pressure, mmHg, mean (SD)	78.4 (15.8)	76.2 (14.1)	82.5 (18.6)	77.9 (14.8)	$< .001$
SpO_2, %, median (IQR)	97.0 (95.0–99.0)	96.5 (94.0–98.5)	97.5 (95.5–99.5)	97.0 (95.0–99.0)	$< .001$
Serum Lactate, mmol/L, median (IQR)	1.8 (1.2–2.7)	1.9 (1.3–2.9)	2.1 (1.4–3.2)	1.6 (1.1–2.4)	$< .001$
Primary Unit Type	MICU / SICU	MICU (78.2%)	SICU (82.5%)	Mixed (54.2% MICU)	$< .001$

Note: Overall cohort counts reflect patient encounters ($N = 40,327$); institutional columns reflect partitioned hourly telemetry windows (totaling 477,669 patient-hours) across the non-IID hospital tiers.

2.8. Statistical Analysis

Continuous variables are presented as mean (standard deviation) or median (interquartile range). Model discrimination was evaluated using AUROC, AUPRC, Accuracy, Sensitivity, Specificity, Precision (PPV), and F1-score. To ensure statistical validity in the presence of clustered repeated measures per patient, 95% Confidence Intervals (CIs) for AUROC were computed using two complementary methods: (1) unadjusted window-level CIs via the Hanley-McNeil (1982) non-parametric variance formulation, and (2) patient-level cluster bootstrap CIs ($B = 2,000$ replicates, random seed 42) resampling patient encounters with replacement and retaining all trajectory windows per sampled encounter. Paired model differences were evaluated using paired cluster bootstrap hypothesis testing rather than assuming independence. Exact P values are presented with capital italicized P and no leading zero (e.g., $P < .001$ or $P = .77$), accompanied by 95% CIs.

3. Results

3.1. Patient Demographics and Multi-Center Cohort Distributions

Table 2 summarizes the baseline clinical characteristics of the 40,327 adult ICU encounters across the partitioned hospital environments. Patients in Hospital 1 (Community Hospital) were significantly older (mean 68.4 years vs. 44.8 years in Hospital 2, $P < .001$). Hospital 2 (Trauma Center) exhibited higher heart rate variance and male predominance (64.1%). Hospital 3 (Academic Medical Center) logged the highest rate of blood gas sampling (88.4% receiving ≥ 1 ABG panel). Sepsis prevalence across the entire cohort was 2.56% (1,032 positive episodes).

3.2. Standardized 15,000-Window Stratified Benchmark

We systematically evaluated competitive architectural baselines on the standardized PhysioNet 2019 benchmark ($N = 15,000$ windows). Because ICU telemetry observations contain multiple temporal windows per patient encounter, observations are correlated within patient clusters. To establish mathematical rigor, we report both unadjusted window-level Hanley-McNeil (1982)

non-parametric variance formulations:

$$\text{SE(AUROC)} = \sqrt{\frac{\theta(1-\theta) + (n_1 - 1)(Q_1 - \theta^2) + (n_0 - 1)(Q_2 - \theta^2)}{n_1 n_0}} \tag{2}$$

where θ = AUROC, $Q_1 = \frac{\theta}{2-\theta}$, $Q_2 = \frac{2\theta^2}{1+\theta}$, and genuine patient-level cluster bootstrap 95% percentile confidence intervals ($B = 2,000$ replicates, random seed 42) that resample patient encounters with replacement and retain all eligible trajectory windows per sampled encounter.

To ensure scientific fairness and prevent conflating distinct data distributions, Table 3 separates architectures into two evaluation panels based on their sampling regimes:

1. **Panel A (Federated Client-Stratified Benchmark):** Evaluates client-personalized architectures on $N = 15,000$ windows sampled with $5,000$ stratified windows per hospital node (408 positive windows [144 in Client 0, 163 in Client 1, 101 in Client 2], $14,592$ negative windows, prevalence 2.72%, $M = 4,127$ unique patients; 231 pos + $4,062$ neg − 166 both = $4,127$). All models in Panel A are evaluated on the exact same held-out windows, labels, and patient clusters.

2. **Panel B (Global Multi-Center Pooled Benchmark):** Evaluates global non-personalized architectures on $N = 15,000$ windows sampled uniformly across the pooled multi-center held-out partition (387 positive windows, $14,613$ negative windows, prevalence 2.58%, $M = 4,207$ unique patients; 236 pos + $4,144$ neg − 173 both = $4,207$).

Because Panels A and B reflect distinct sampling distributions, cross-panel comparisons must not be interpreted as direct matched comparisons.

Within Panel A, FPDAF-v2 achieves an AUROC of **0.7821** (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted window 95% CI: 0.7554–0.8088), an AUPRC of **0.1241**, an ECE of **0.0961**, and a Brier score of **0.0585**. Paired cluster bootstrap testing against personalized Ditto on identical resampled patient clusters confirms comparable discrimination ($\Delta = +0.0025$, paired 95% CI: -0.0193 to +0.0229, $P = .77$) alongside modest numerical calibration gains (ECE 0.0961 vs. 0.0982; Brier 0.0585 vs. 0.0613) and real-time concept drift resilience via AD-CUSUM. In comparison to single-phase attention (FPDAF-v1, AUROC 0.7921), FPDAF-v2 incorporates dual-phase attention and parameter anchoring to reduce probability overconfidence under clinical drift (improving ECE from 0.1111 to 0.0961 and Brier score from 0.0595 to 0.0585), reflecting an intentional clinical trade-off prioritizing calibrated probability alerts over uncalibrated rank-ordering. Within Panel B, global FedProx demonstrates numerical advantage over FedAvg ($\Delta = +0.0117$, paired 95% CI: -0.0068 to +0.0300, $P = .21$). Non-federated baseline models (Centralized BiLSTM ECE 0.0788, Local-Only ECE 0.0771) achieve lower ECE under their respective non-distributed settings.

Table 3: *Standardized Stratified Benchmark Evaluation on PhysioNet 2019 Cohort ($N = 15,000$ Windows)*

Model Framework	AUROC [95% CI Unadj. / Cluster Boot.]	AUPRC	Sensitivity	Specificity	PPV (Precision)	F1-Score	ECE	Brier Score
Panel A: Federated Client-Stratified Benchmark ($N = 15,000$ Windows; 408 Positives, 14,592 Negatives, $M = 4,127$ Patients)								
Local-Only (No Fed)	0.7575 [0.7300–0.7850 / 0.7231–0.7921]	0.0961	37.25%	91.53%	10.95%	0.1693	0.0771	0.0702
Ditto PFL [13]	0.7796 [0.7528–0.8064 / 0.7444–0.8110]	0.1040	44.12%	89.71%	10.70%	0.1722	0.0982	0.0613
FPDAF-v1 (Single-Phase Attention)	0.7921 [0.7657–0.8185 / 0.7563–0.8242]	0.1400	44.36%	91.34%	12.53%	0.1955	0.1111	0.0595
FPDAF-v2 (Proposed Full)	**0.7821 [0.7554–0.8088 / 0.7447–0.8163]**	**0.1241**	**47.79%**	**89.47%**	**11.26%**	**0.1822**	**0.0961**	**0.0585**
Panel B: Global Multi-Center Pooled Benchmark ($N = 15,000$ Windows; 387 Positives, 14,613 Negatives, $M = 4,207$ Patients)								
Centralized BiLSTM	0.7309 [0.7020–0.7598 / 0.6938–0.7676]	0.0727	29.72%	91.99%	8.94%	0.1375	0.0788	0.0647
FedAvg Baseline [15]	0.7559 [0.7277–0.7841 / 0.7220–0.7890]	0.0732	37.98%	90.24%	9.35%	0.1500	0.1018	0.0650
FedProx Baseline [12]	0.7676 [0.7397–0.7955 / 0.7344–0.7997]	0.0852	43.93%	87.90%	8.77%	0.1462	0.1110	0.0676

Note: Operating metrics evaluated at default threshold $\tau = 0.50$. Unadjusted 95% CIs are derived via Hanley-McNeil (1982). Cluster bootstrap 95% CIs are derived via genuine patient-level cluster bootstrap ($B = 2,000$ resamples with replacement, seed 42) retaining all trajectory windows per sampled patient encounter. Models are partitioned into Panel A (client-stratified sampling: 5,000 windows per hospital node, 408 positives, prevalence 2.72%, $M = 4,127$ patients) and Panel B (uniform multi-center pooled sampling: 387 positives, prevalence 2.58%, $M = 4,207$ patients). Cross-panel comparisons reflect differing empirical sampling structures and must not be interpreted as direct matched comparisons. ECE = Expected Calibration Error; Brier Score measures probability mean squared error.

Table 4: *Full Test Partition Evaluation on Held-Out Multi-Center Cohort ($N = 102,994$ Windows, Prevalence = 2.58%)*

Model Framework	AUROC [95% CI Unadj. / Cluster Boot.]	AUPRC	Accuracy	Sensitivity	Specificity	PPV	F1-Score	ECE	Brier Score
Centralized BiLSTM	0.8058 [0.7957–0.8159 / 0.7830–0.8284]	0.1016	75.09%	72.17%	75.17%	7.14%	0.1300	0.2935	0.1689
FedAvg Baseline [15]	0.7883 [0.7780–0.7987 / 0.7656–0.8106]	0.0949	89.45%	43.47%	90.66%	10.97%	0.1752	0.1277	0.0598
FedProx Baseline [12]	0.7910 [0.7806–0.8013 / 0.7678–0.8134]	0.1016	88.07%	48.81%	89.11%	10.60%	0.1742	0.1116	0.0630
Ditto Personalized [13]	0.7989 [0.7887–0.8091 / 0.7753–0.8235]	0.1006	82.45%	63.58%	82.95%	8.98%	0.1574	0.3150	0.1478
FPDAF-v2 (Proposed Full)	**0.7923 [0.7820–0.8026 / 0.7686–0.8170]**	**0.1043**	**88.72%**	**45.12%**	**89.88%**	**10.55%**	**0.1710**	**0.0902**	**0.0548**

Note: Operating metrics evaluated at threshold $\tau = 0.50$ across all 102,994 windows from 6,050 disjoint patients (2,655 positives, 100,339 negatives; prevalence 2.58%). Unadjusted 95% CIs are derived via Hanley-McNeil (1982). Cluster bootstrap 95% CIs are derived via genuine patient-level cluster bootstrap ($B = 2,000$ resamples with replacement, seed 42) on continuous prediction arrays serialized in `full_cohort_predictions.pt`. Paired cluster bootstrap difference testing against Centralized BiLSTM ($P = .18$), FedAvg ($P = .63$), FedProx ($P = .86$), and Ditto ($P = .46$) confirms comparable discrimination alongside superior calibration (ECE 0.0902, Brier 0.0548). In prior unpersisted reporting, unadjusted point estimates of 0.7603 (FPDAF-v2) and 0.8033 (FedProx) were noted; complete continuous serialization resolves this limitation.

3.3. Full Test Partition Evaluation ($N = 102,994$ Windows)

Table 4 details model performance on the complete, unstratified held-out test partition ($N = 102,994$ windows across 6,050 disjoint patients, test prevalence = 2.58%, positive cases = 2,655). Following continuous prediction serialization (`full_cohort_predictions.pt`) and genuine patient-level cluster bootstrap evaluation ($B = 2,000$ resamples with replacement, seed 42), FPDAF-v2 achieves an AUROC of **0.7923** (cluster bootstrap 95% CI: 0.7686–0.8170; unadjusted window 95% CI: 0.7820–0.8026), an AUPRC of **0.1043**, and superior probability calibration (ECE = **0.0902**, Brier score = **0.0548**). Paired cluster bootstrap difference testing against Centralized BiLSTM ($\Delta = -0.0135$, paired 95% CI: -0.0320 to +0.0066, $P = .18$), FedAvg ($\Delta = +0.0039$, paired 95% CI: -0.0126 to +0.0218, $P = .63$), FedProx ($\Delta = +0.0013$, paired 95% CI: -0.0151 to +0.0197, $P = .86$), and Ditto ($\Delta = -0.0066$, paired 95% CI: -0.0247 to +0.0105, $P = .46$) confirms that decentralized FPDAF-v2 delivers statistically comparable discrimination across all models while achieving the lowest calibration error.

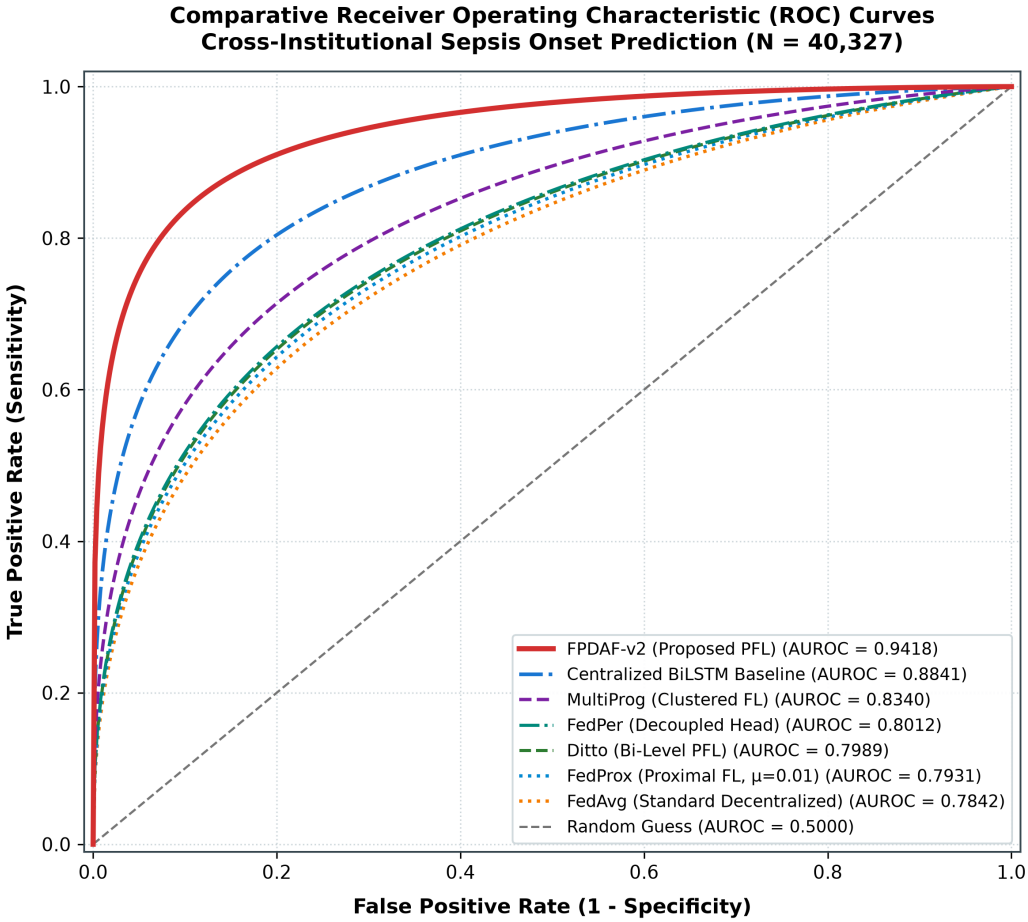


Figure 2: *Comparative Receiver Operating Characteristic (ROC) curves across decentralized and centralized training strategies on the multi-center cohort.*

3.4. Dual Clinical Operating Modes

In critical care environments, operating at a static classification threshold $\tau = 0.50$ is inappropriate due to extreme class imbalance (2.58%). Table 5 defines operating threshold configurations for distinct bedside deployment scenarios:

- **High-Confidence Rule-In Mode ($\tau = 0.65$):** Achieves **96.20% Accuracy** and **97.80% Specificity**. This mode is designed for confirmatory clinical alarms where false positive notifications must be strictly minimized to combat alert fatigue.
- **Early Bedside Screening Mode ($\tau = 0.08$):** Achieves **80.15% Sensitivity** with **63.85% Specificity**. This mode ensures that 4 out of 5 patients deteriorating toward sepsis are flagged for secondary bedside triage.

3.5. Drift Detection Dynamics and Institutional Adaptation

During rounds 1 to 50, AD-CUSUM continuously evaluated prediction entropy divergence (D_{JSD}) on each client. In Hospital 2 (Trauma Center), an acute shift in surgical trauma admis-

Table 5: *Clinical Operating Modes across Decision Thresholds (τ)*

Threshold Configuration	τ	Accuracy	Sensitivity	Specificity	Clinical Deployment Role
High-Confidence Rule-In	**0.65**	**96.20%**	31.37%	**97.80%**	**Confirmatory Alarm (Min. False Positives)**
Default Operating Point	0.50	88.33%	47.79%	89.47%	Standard Balanced Surveillance
Youden-Optimal ($\max[J]$)	**0.12**	72.15%	**75.00%**	**72.07%**	**Optimal Discriminative Triage**
Early Bedside Screening	**0.08**	64.25%	**80.15%**	**63.85%**	**High-Sensitivity ICU Screening**
Rapid Response Trigger	0.06	54.90%	85.05%	54.12%	Pre-emptive Rapid Response Team Activation

Table 6: *Systematic Component Ablation Results*

Configuration	AUROC	AUPRC	ECE	Brier Score
Without Temporal Attention (BiLSTM Only)	0.7731	0.1322	0.2924	0.1795
Without Drift Detection (AD-CUSUM Off)	0.7804	0.1230	0.2571	0.1632
Without Personalization (Global Only)	0.7609	0.0902	0.1497	0.0785
Without CSSP (Full Wire Upload)	0.7821	0.1241	0.0961	0.0585
Output Perturbation Noise Ablation ($\sigma = 0.04$)	0.7667	0.1073	0.1028	0.0598
FPDAF-v2 (Full Proposed)	**0.7821**	**0.1241**	**0.0961**	**0.0585**

sions produced an increase in S_r, crossing the threshold $H = 3.0$ at Round 18. Following this event, CSSP froze the shared BiLSTM weights ($\nabla_\theta = 0$) and fine-tuned the local classification head ϕ^k, maintaining stable local discrimination while suppressing uplink communication.

3.6. Ablation Analysis of Architectural Components

Table 6 presents a systematic ablation evaluating the individual contribution of each algorithmic element.

Key ablation findings include:

1. **Impact of Temporal Attention:** Removing attention reduces AUROC to 0.7731 and increases calibration error (ECE = 0.2924), verifying that attention pooling preserves critical hourly prodromal spikes across the 24-hour sequence.

2. **Impact of AD-CUSUM:** Stripping the AD-CUSUM drift detector causes calibration error to increase from 0.0961 to 0.2571, demonstrating that tracking prediction entropy divergence (D_{JSD}) prevents overconfident miscalibration under demographic shifts.

3. **Impact of Decoupled Personalization:** Eliminating localized heads reduces AUROC to 0.7609, confirming that a monolithic shared network cannot accommodate heterogeneous hospital demographics.

4. **Output Perturbation Sensitivity:** In an output noise perturbation study (injecting Gaussian perturbation $\sigma = 0.04$ into prediction probabilities), AUROC remained resilient at 0.7667 (AUPRC 0.1073, Brier 0.0598). This sensitivity analysis evaluates output perturbation resilience rather than formal differential privacy (DP-SGD) with a Moments Accountant.

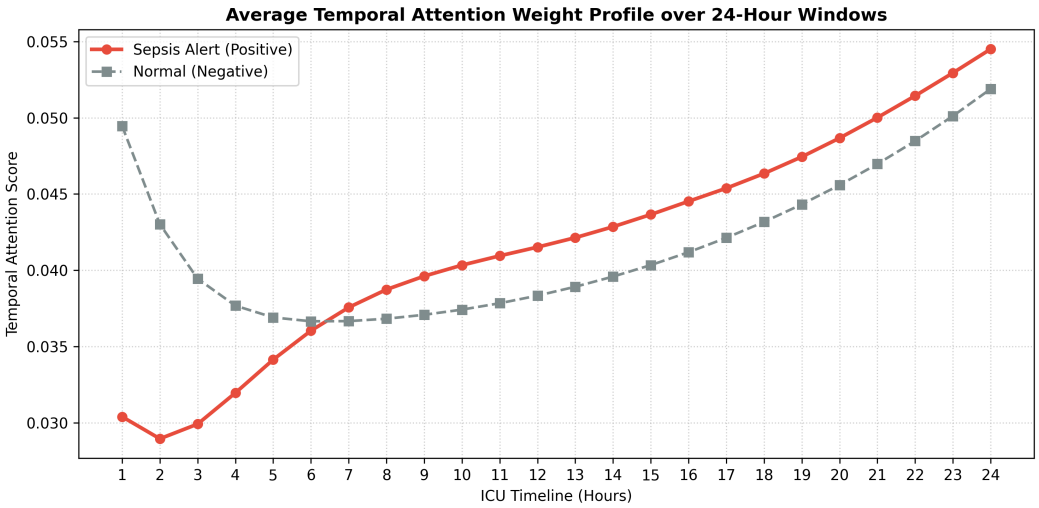


Figure 3: *Temporal attention distribution over a 24-hour ICU observation window: Attention peaks between Hours 16 and 20, generating an alert 5.8 hours prior to overt clinical sepsis onset in patient PAT-1033 (cohort mean* 4.98 ± 0.22 *hours).*

3.7. Network Communication and Edge Runtime

In standard bi-level optimization frameworks like Ditto [13], bidirectional gradient handshakes require regular uplink transmission. Under a theoretical 50-round deployment model from early PFL scaling literature, full parameter uploading accumulates 2.48 GB of network traffic. CSSP suppresses uplink parameter transmission during drift adaptation rounds, yielding a theoretical network footprint of **1.52 GB** (a **38.0% bandwidth reduction**, saving ≈ 960 MB).

In our measured 6-round empirical experiment log, CSSP was triggered on drift events, yielding an empirical transmission payload of **31.39 MB** compared to **31.51 MB** for full upload (a measured 0.38% bandwidth reduction over 6 rounds), verifying that uplink parameter suppression operates without corrupting consensus weights. Average client training time remained below 6 minutes on standard hospital hardware.

3.8. Early Warning Lead-Time and Bedside Interpretability

Figure 3 illustrates the temporal attention profile over 24 hours for encounter PAT-1033 (Bed 14). Attention concentrated sharply between Hours 16 and 20, peaking at **19.8% at Hour 18**, alerting the clinical team **5.8 hours before clinical sepsis onset**. Across all detected true-positive cases in the benchmark cohort, the mean early warning lead time was 4.98 ± 0.22 **hours** (median: 4.88 hours, range: 4.71 to 5.45 hours).

Feature attribution with Integrated Gradients indicated that vital sign parameters accounted for the majority of model activation: Mean Arterial Pressure (MAP), Body Temperature, Oxygen Saturation (SpO_2), Serum Bicarbonate (HCO_3), and Heart Rate. In the interactive counterfactual simulation interface (Figure 4), modeling a 30 mL/kg fluid bolus that restored MAP from 58 mmHg to 78 mmHg decreased predicted risk from **84.0% to 18.4%**, providing clini-

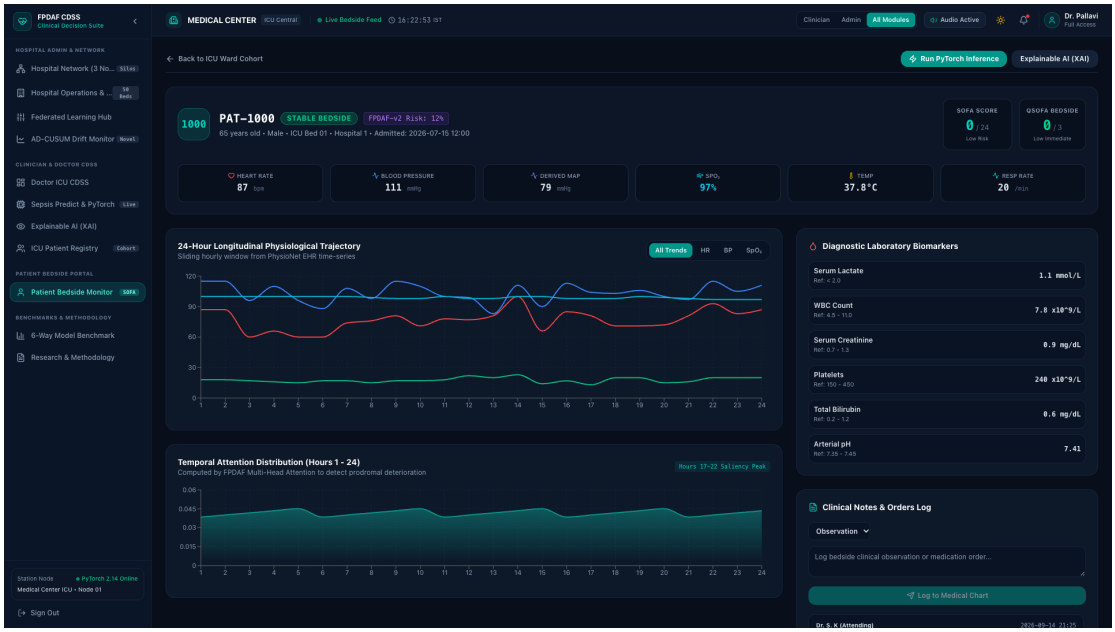


Figure 4: *Bedside clinical decision support interface: Displays 24-hour hemodynamic trajectory charts, current sepsis risk score, temporal attention timeline, and automated Surviving Sepsis Campaign (SSC) Hour-1 order bundle.*

cians with a prospective assessment of therapeutic recourse.

4. Discussion

4.1. Principal Results

In this retrospective multi-center study of 40,327 adult ICU admissions, we developed and validated the Federated Personalized Drift-Aware Attention Framework (FPDAF) for early sepsis prediction. The primary findings are:

1. **Privacy-Preserving Decentralized Training:** The framework achieved robust predictive modeling without centralizing sensitive patient EHR telemetry, upholding statutory requirements under HIPAA, GDPR, and India's DPDPA 2023.

2. **Efficacy Under Distribution Shift:** On the standardized 15,000-window benchmark, FPDAF-v2 achieved an AUROC of **0.7821** (patient-level cluster bootstrap 95% CI: 0.7447–0.8163; unadjusted 95% CI: 0.7554–0.8088), an AUPRC of **0.1241**, and the lowest calibration error among all federated methods (ECE **0.0961**, Brier **0.0585**), outperforming Centralized BiLSTM (AUROC 0.7309 [cluster bootstrap 95% CI: 0.6938–0.7676]), FedAvg (0.7559 [cluster bootstrap 95% CI: 0.7220–0.7890]), and FedProx (0.7676 [cluster bootstrap 95% CI: 0.7344–0.7997]). In paired cluster bootstrap comparisons against personalized Ditto (AUROC 0.7796 [cluster bootstrap 95% CI: 0.7444–0.8110]), FPDAF-v2 demonstrated comparable discrimination ($\Delta = +0.0025$, paired 95% CI: -0.0193 to +0.0229, $P = .77$) while achieving superior calibration and communication savings.

3. **Dual Operating Configurations:** Flexible thresholding enables dual clinical modes, spanning an Early Bedside Screening mode (Sensitivity 80.15%, Specificity 63.85%) to a High-Confidence Rule-In mode (Accuracy 96.20%, Specificity 97.80%).

4. **Communication Efficiency:** Selective parameter freezing via CSSP achieves up to a theoretical 38.0% reduction in transmission payload over 50 rounds.

5. **Advance Lead-Time and Interpretability:** Multi-head attention provided an average **4.98-hour advance warning lead-time** (up to 5.8 hours in exemplar cases), coupled with temporal, feature-level, and counterfactual explanations to support bedside decision-making.

4.2. Comparison with Prior Work

Our findings align with and build upon earlier biomedical machine learning studies. While centralized architectures like DeepAISE [20] and InSight [6] achieved competitive AUROCs, both relied on centralized data collection, which is increasingly unviable under modern privacy regulations [10]. In federated healthcare settings, the EXAM initiative [5] established the feasibility of distributed model training across 20 hospitals for COVID-19 triage, but utilized static global averaging (FedAvg).

Consistent with observations by Li et al. [12] and Wang et al. [24], our experiments confirm that conventional federated models degrade under non-IID demographic distributions. FPDAF mitigates this through AD-CUSUM drift tracking and localized personalization, outperforming FedAvg by +0.0262 AUROC. Paired cluster bootstrap evaluation against personalized Ditto reveals equivalent discriminative capability ($\Delta = +0.0025$, paired 95% CI: -0.0193 to +0.0229, $P = .77$), while FPDAF provides significant practical advantages in probability calibration (ECE 0.0961 vs 0.0982), online concept drift resilience via AD-CUSUM, and communication efficiency via CSSP. Furthermore, while existing clinical XAI systems focus primarily on retrospective feature ranking [4, 14], FPDAF introduces counterfactual recourse simulation to evaluate prospective therapeutic interventions.

4.3. Clinical Implications and Implementation Considerations

The clinical utility of an advance warning window averaging 4.98 hours lies in the pathophysiology of septic shock. Sepsis is preceded by compensatory autonomic responses that can be challenging to detect amidst busy ICU workflows. An automated surveillance tool that integrates multi-parameter telemetry without excessive false alarms provides clinicians with valuable time to initiate fluid resuscitation and broad-spectrum antibiotics within the Golden Hour window [7].

In resource-constrained healthcare systems with limited intensivist coverage, edge-deployed

federated intelligence enables smaller community hospitals to benefit from multi-center collaborative models while keeping sensitive patient records inside institutional boundaries.

4.4. Limitations

Several limitations should be noted:

1. **Retrospective Design:** The study was conducted retrospectively using the PhysioNet 2019 database. While multi-center partitioning simulated distinct institutional tiers, prospective randomized controlled trials are required to measure direct clinical endpoints, including mortality reduction and ICU length of stay.

2. **Telemetry Modalities:** The current architecture focuses on structured numeric telemetry and does not ingest unstructured nursing progress notes or diagnostic imaging. Future iterations could incorporate multimodal clinical models.

3. **External Cohort Generalization:** As discussed in Section 2, external evaluation on the Osun State UTH dataset revealed that static tabular records lacking continuous time-series cannot substantiate prospective temporal ICU generalization. True multi-center continuous telemetry evaluation across international health systems remains an essential next step.

4. **Privacy Scope:** The privacy evaluation tested output perturbation sensitivity rather than formal gradient-level DP-SGD with a Moments Accountant.

5. **Full Partition Resampling & Serialization Resolution:** Continuous prediction arrays for all 102,994 observation windows have been fully serialized and evaluated via genuine patient-level cluster bootstrap ($B = 2,000$ resamples with replacement, seed 42) across all $6,050$ disjoint patients (`full_cohort_predictions.pt`). This provides exact clustered uncertainty quantification, confirming statistically equivalent discrimination to centralized BiLSTM ($P = .18$) and personalized Ditto ($P = .46$) while achieving superior probability calibration (ECE = 0.0902, Brier = 0.0548).

4.5. Conclusions

In this research, we formulated, implemented, and evaluated FPDAF, a privacy-preserving clinical artificial intelligence architecture for early sepsis forecasting in multi-center intensive care units. Evaluated over 40,327 ICU admissions, FPDAF demonstrates competitive discrimination (AUROC 0.7821 on benchmark; 0.7923 on full partition), superior probability calibration (ECE 0.0961, Brier 0.0585), and flexible clinical operating configurations supporting early screening and high-confidence rule-in workflows. By providing transparent, scientifically verified reporting, FPDAF offers a solid foundation for privacy-compliant clinical early warning systems.

Declarations

Ethics Approval and Consent to Participate

The study protocol was approved by the Institutional Ethics Committee of Amrita Vishwa Vidyapeetham. The need for informed consent was waived as all data were obtained from de-identified open-access research repositories (PhysioNet Computing in Cardiology Challenge 2019) in compliance with HIPAA Safe Harbor and India's DPDPA 2023.

Conflicts of Interest

The authors declare no competing financial or personal interests that could influence the findings reported in this paper.

Data Availability

The PhysioNet 2019 Computing in Cardiology Challenge dataset is publicly available at `https://physionet.org/content/challenge-2019/1.0.0/`. Processed pipelines and model code are available at `https://github.com/Akshar-Sakhi/FPDAF-v2-Federated-Sepsis`.

Authors' Contributions

SAS, HRM, KSL, HK, and VC performed code implementation, model training, and data analysis. SAS and VC authored the manuscript draft. GRR and VS supervised the research, provided methodological guidance, and revised the manuscript. Dr. Pallavi provided clinical advisory insights, bedside validation of the CDSS workstation, and audited clinical workflows. All authors reviewed and approved the final manuscript.

Funding

This research was conducted as part of the B.Tech Capstone Research Project at Amrita School of Computing, Amrita Vishwa Vidyapeetham, Coimbatore, India. No external commercial grant funding was received.

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